03 · Price elasticity by region — random slopes & endogeneity

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The decision. We're setting regional prices. Elasticity — how much demand moves when we change price — differs by region, but two traps lurk: (1) small regions give **noisy** estimates we shouldn't over-trust, and (2) **price is endogenous** — we historically cut prices *because* demand was soft, so a naive regression understates elasticity.

We estimate a **partially-pooled random-slope** elasticity per region (noisy regions borrow strength from the fleet), turn each into a **profit-maximising price** with uncertainty, and then show honestly what partial pooling **cannot** fix — the endogeneity — quantifying the bias and previewing the instrumental-variables cure (notebook 11).

> Cookbook maps this to `pathmc` random slopes; here we fit the equivalent hierarchical model explicitly in PyMC so the shrinkage is fully visible. The identification logic is unchanged.

1.1 2 · Simulate a ground truth

A region x week panel with a **constant-elasticity (log-log) demand curve** $\log Q_{rt} = \alpha_r + \beta_r \log P_{rt} + \gamma^\top Z_{rt} + \varepsilon_{rt}$, where β_r is region r's true price elasticity (planted, varying around a fleet mean ~ -1.4). Observed confounders Z = competitor price, seasonality, trend. We start with **no** endogeneity so we can show recovery, then switch it on in step 7.

```
12 regions, fleet-mean elasticity -1.42; obs per region range 16-80 (regions 0-2
are thin, to exercise shrinkage)

      region  week       price    demand  competitor_price        season  \
0  region_00     2  13.003239  0.159630         20.405765  1.394170e-01
1  region_00     3  14.245949  0.157405         16.535730  1.989368e-01
2  region_00    12  14.310937  0.159097         20.819276  7.179470e-02
3  region_00    13  15.626376  0.175255         21.659711 -9.648736e-17
4  region_00    15  13.204126  0.152195         19.486540 -1.394170e-01

   trend  log_price  log_demand
0   0.02   2.565199   -1.834900
1   0.03   2.656473   -1.848934
2   0.12   2.661024   -1.838239
3   0.13   2.748960   -1.741513
4   0.15   2.580529   -1.882592
```

1.2 3 · Identify — elasticity as a random slope, and the backdoor set

Estimand. Per-region elasticity $\beta_r = \partial \log Q / \partial \log P$. **Partial pooling** treats $\beta_r \sim \mathcal{N}(\mu_\beta, \tau_\beta^2)$ — noisy regions shrink toward the fleet mean μ_β, and τ_β *measures how much elasticity genuinely varies*. The **shrinkage factor** for region r is approximately $1 - \dfrac{\tau_\beta^2}{\tau_\beta^2 + \sigma^2/n_r}$ — thin regions (small n_r) get pulled harder toward the fleet. Shrinkage is a feature: it stops us over-reacting to a small region's noisy slope.

Confounding is the crux. Price is endogenous — a demand shock moves both P and Q. Conditioning on the observed backdoor set (competitor price, seasonality, trend) closes the *observable* paths. What pooling does **not** fix is an *unobserved* common shock — for that you need an instrument (notebook 11). We state this out loud rather than hide it.

1.3 4 · Estimate — three pooling regimes, side by side

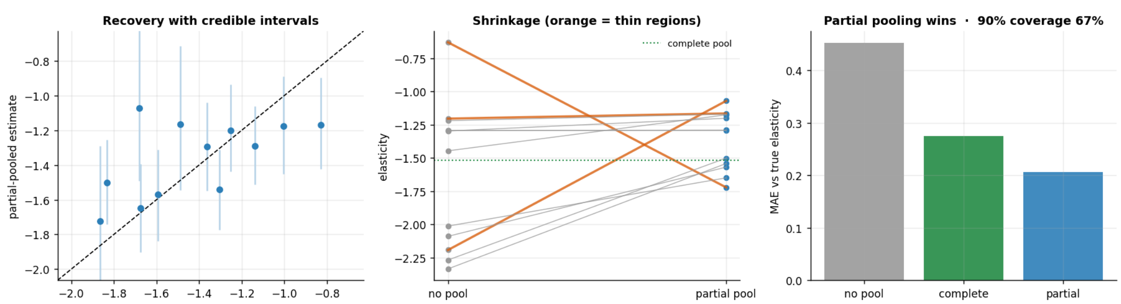
We fit the same elasticity three ways to make the bias–variance trade-off concrete: **no pooling** (per-region OLS — unbiased but noisy), **complete pooling** (one fleet elasticity — stable but ignores real variation), and **partial pooling** (the hierarchical Bayes middle ground). Only partial pooling gets both the fleet level and the region-to-region spread right.

```
fleet-mean elasticity mu_beta -1.36 (true -1.42)  ·  between-region spread
tau_beta 0.31
```

1.4 5 · Validate — recovery, shrinkage, and calibration

Estimated vs true elasticity should track the 45° line; the thin regions' point estimates get pulled toward the fleet mean (that's pooling protecting us), and the credible intervals are wider where data is thin. We also check that the 90% intervals actually **cover** the true elasticities (calibration).

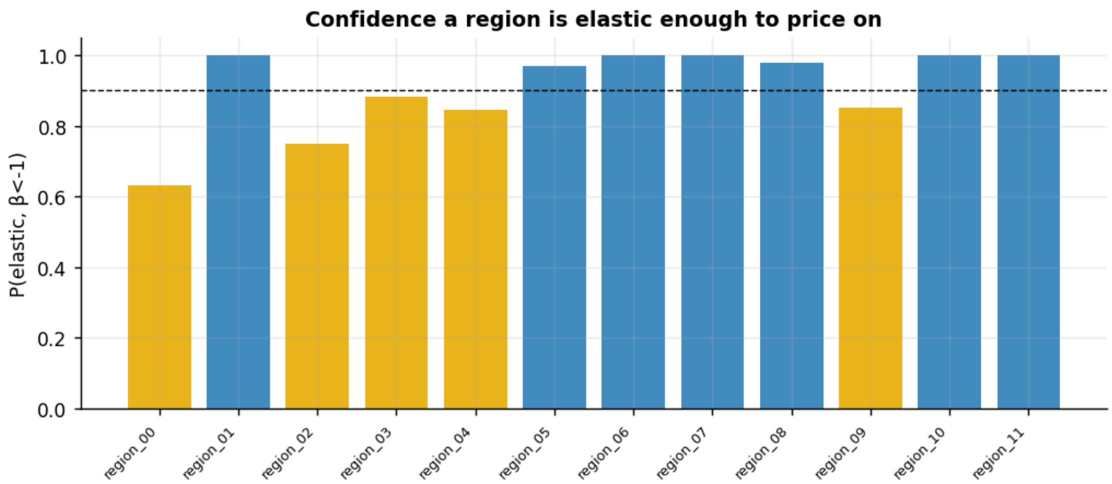
```
MAE - no pool 0.45, complete 0.28, partial 0.21  ·  coverage 67%
```



1.5 6 · Decide, in euros — profit-maximising price per region

For a constant-elasticity demand curve with marginal cost c, the profit-maximising price is $P^\star = \dfrac{\beta}{\beta+1}c$ (needs $\beta < -1$). We propagate the **elasticity posterior** into a **price posterior** per region, and — the honest part — flag regions where we're not confident demand is even elastic ($\beta < -1$) for a **controlled price test** rather than acting on a shaky slope.

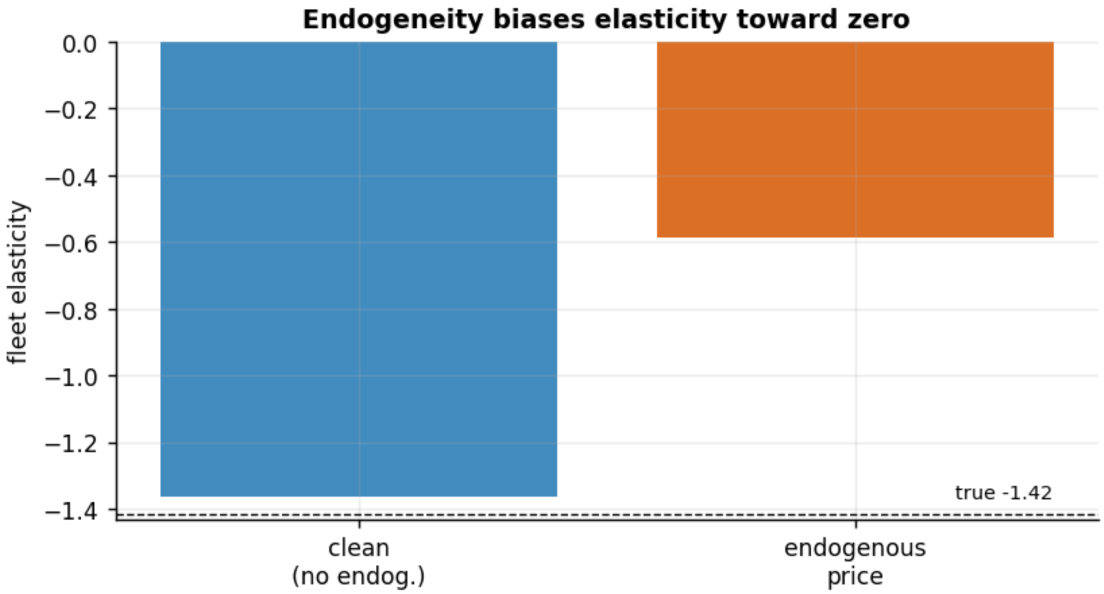
region	elasticity	P(elastic)	opt_price	action
region_00	-1.07	0.63	€43.4	controlled test
region_01	-1.72	1.00	€19.6	set price
region_02	-1.17	0.75	€38.4	controlled test
region_03	-1.20	0.88	€41.7	controlled test
region_04	-1.18	0.85	€45.5	controlled test
region_05	-1.29	0.97	€34.4	set price
region_06	-1.57	1.00	€22.2	set price
region_07	-1.65	1.00	€20.3	set price
region_08	-1.29	0.98	€35.1	set price
region_09	-1.17	0.85	€47.0	controlled test
region_10	-1.54	1.00	€22.7	set price
region_11	-1.50	1.00	€24.0	set price



1.6 7 · Caveats — what partial pooling does *not* fix

Partial pooling handles *noise*, not *bias*. If price is endogenous to an **unobserved** demand shock, adjusting for observed Z isn't enough and elasticity is biased **toward zero**. We demonstrate by switching the hidden shock on — and preview the IV cure.

```
clean fleet elasticity -1.36 (true -1.42)
endogenous fleet elasticity -0.59 (true -1.42) -> biased toward zero by +0.83
Fix = an instrument (a cost shifter) or a natural price experiment - see
notebook 11 (IV).
```



Other caveats. Log-log assumes *constant* elasticity within region (a local approximation if elasticity varies with price level); `lag(price)` would capture stockpiling dynamics (omitted for clarity); and shrinkage is a prior choice — too few regions or a mis-specified τ_β prior can over-pool, so always inspect the fleet-vs-region spread.